



Lore Document: Reactor Core and Unknown Technologies

The Reactor Core

The reactor core itself is built out of a material called **AD182 plasma**, which is extremely unstable and wants to let out all of its energy immediately, which makes it extremely dangerous

Alone, AD182 plasma is generally safe This reaction only happens when it comes into contact with even **0.2% of AD183**

If this reaction happens on a mass scale, it will produce the equivalent of **321 20,000-kiloton nuclear bombs** in less than 0.1 seconds, along with irradiating everything around it and causing mass contamination and destruction

Unknown Technologies Corporation

Unknown Technologies Corporation was a research organization founded in 1971, devoted to developing clean and safe energy

The first major project by the organization was the **Rutherford Generation Site**, which was created in the 1970s. The Rutherford Generation Site contained the first major achievement, which was the **Cold Fusion Reactor (CFR)** The major point of the technology, along with the reactor design, was designed by **Dr. Ryker**, but he was fired

The Rutherford Generation Site became operational in the 1980s and was a massive success, as they were successfully able to create a fusion reactor and sync it with the grid

This site remained operational all the way until 2015, when it was shut down and replaced with a newer site  because of age

In 2071, 5 years after site alpha was destroyed, **UTPRC Site Beta** started construction and was finished in 2091. They also used equipment from another corporation, **EEI** such as the reactor core. Currently, the site is ran by 2 people,   (Codename: Astral) And Vex

All other personnel live offsite.

UT Site Alpha

Due to the nature of site Alpha's destruction, all documents remain **STRICTLY** classified about the ongoings of Alphas research.

All surviving Personnel have been terminated or, reassigned to **UT SITE BETA**

Discovery of AD182 and AD183

In 2018, researchers in the Unknown Technology Site [REDACTED], down in Research Lab 15WA, discovered they could create this so-called "material," which was named **AD182**. It could be easily created with minimal energy. It is a type of "plasma," if you would call it that

The researchers did not think much of it until they found a similar material named **AD183**. When combined, it was found that they would release an extreme amount of energy very quickly, seemingly from nowhere. This reaction would continue until the two materials were separated

Unknowingly, they combined both of these elements, which completely obliterated Research Lab 15WA. After that incident, more precautions were implemented

Researchers theorized that if they could somehow contain these two materials and keep them together, they could generate massive amounts of power

Shield Development

After researching ways to contain this, they developed a type of **shield** which almost acted as a force field. It could be shaped in many different ways and was incredibly strong, resistant to basically anything. It was also extremely light, weighing close to nothing, which made it easily suspended midair

They developed a small experimental "reactor" which was no bigger than a microwave. They also developed specialized **power lasers** to deliver AD182 and AD183 into a particular point suspended midair with the shield they developed, which could be formed using similar power lasers labeled **stabilization lasers**

This reactor, after **5,658 variations**, was a success. The first time it worked, they had it on for only 5 seconds before it generated so much heat they had to turn it off

Another discovery was made during testing. The reaction was the only possible thing they could find that would destroy the shield. Although it was noted that the shield would be a viable option if they could figure out how to slow the reaction

SPF192 Discovery

Around the 2040s, they found the game changer. They found a new material called **SPF192**, which had the effect of slowing the reaction. After more studies, they concluded that with enough SPF192, they could effectively stop the reaction, destroying both elements in the process while releasing the energy slowly instead of in an extremely fast burst

After adding this limiter to their test reactor setup, they managed to maintain a reaction seemingly indefinitely. It only stopped when they ran out of one of the materials. It was determined after full testing that this reaction output **256,000 times more energy than was added**

After doing more tests ensuring that this could be scaled up, funding was acquired to create a new research site which was named **UT Site Beta**

UT Site Beta and the IPRCMK1 Reactor

Unknown Technology Site Beta began construction in 2071 and became operational in 2091 (although strong rumors within the company stated that it began construction in 2081, which has been proven false). In Sector C of the new research complex, they started construction on a massive scaled-up version of their tests. The reactor was called the **IPRCMK1**, and this had to be a success, otherwise everything could be destroyed.

Once the reactor construction was completed, its safety factor was estimated and determined that, assuming everything was fine-tuned perfectly, the startup would have a disastrous failure chance of **0.02%**. Even though rare, the technicians installed a special panel in the control room called the **Startup Abort Button**, which would instantly dump half the supply of SPF192 into the reactor immediately (**2981% more than what is required on a normal shutdown**).

The startup was also estimated to have a non-disastrous failure chance of **10%**, where the reactor's onboard **RCC computer** would immediately detect something was wrong and would automatically shut it down within **1.2 picoseconds**

Reactor Configuration

- **6 Power Lasers** – Deliver and control AD182 and AD183
- **2 Stabilization Lasers** – Create the shield and control SPF192
- **2 Power Laser Booster Controls**
- **2 Coolant Controls**
- **6 Vent Fan Controls**
- **2 Alternate Vent Controls**

If the stabilization lasers were to fail and the RCC failed to restart them, they were designed to retract into their holding bay as a precaution to ensure they were not damaged by a possible shield explosion

If the stabilization lasers were to fail, it was deemed that the shield could last 25 minutes before exploding. Although short in the grand scheme, the reactor cannot be shut down without them.

The stabilization lasers' failure chance per hour was estimated to be around **0.0000007284%**, well within safe ranges.

The reactor was equipped with monitoring systems to monitor every bit of the reactor.

The reactor startup system was protected by a startup request system. Reactor operators had to request startup from HQ in order for startup to begin.

A controversial addition was made after people kept shutting down the reactor unexpectedly and causing problems around the facility. Verification panels were installed.

Before the emergency shutdown system could commence, each operator had to input a code and verify. The reason for implementing this is unclear and raised many safety concerns. Some personnel quit the facility entirely because of this.

Reactor Operations

In the event of **shield degradation**, which was estimated to happen around **4,000K**, reactor personnel were instructed to cool the reactor down manually if the temperature was not above **4,500K** and integrity was not below **80%**. Otherwise, they were instructed to activate the **emergency coolant reserve**.

The reactor was constantly supplied with **SPF919** to ensure it did not explode.

Protocols

PROTOCOL DELTA

Developed by Section ██████ in the ALEPH Control Wing. Its purpose was to ██████. It can be activated in the ██████ section by ██████ and needs ██████ before activation.

PJT Solarity

Can be accessed in ██████. PROJECT SOLARITY contains ██████, which can be used in the case of ██████, OSHA ██████, or ██████. Its current effect of ██████ has been estimated to be around ██████. The ██████ takes around 5 minutes to ██████.

PROTOCOL NULLWAVE

This protocol was developed in case the shield became critically unstable. It prepares the reactor for shutdown after the shield has already failed using highly advanced methods to neutralize the reaction enough to become harmless without the shield.

PROTOCOL BETA

N/A – Not canon

PROTOCOL NEXIS

Protocol NEXIS is an emergency protocol designed to limit the disastrous effects of the reactor KV rising to extremely dangerous levels. It is designed to optimize the reactor in ways that lower KV and reduce the potential damage caused by a runaway reactor. This protocol sets evacuation orders and also limits destruction (but does not prevent it).

Meltdown Shutdown

In order to shut down the reactor during a meltdown, it is much harder as there is no shield to contain the SPF912. It estimates whether the stabilization lasers were damaged, determines whether the reactor KV is too high, and how long the meltdown has been ongoing.

The reactor temperature was observed to have a massive effect on the KV levels along with the overall success chance of the shutdown. This was learned in **Incident Report #D8284**, where the reactor operator (Tim) fell asleep and was awoken from the shield exploding. All protocols

were met, and the reactor was able to be shut down within 60 seconds Tim was fired after this and, using the **GDSFT**, was successfully sent back Tim does not remember us or anything about the facility

GDSFTMK1

A **time machine and teleporter** working by the principles of [REDACTED]

Incident Reports

Incident Report #28164

After the reactor operations personnel stalled the reactor six different times in the same day, the shift manager Bob, along with the entire reactor operations team, was fired

It is also noted that the security teams failed to respond to the A1 call by [REDACTED] and were reprimanded

Research Lab 15WD Imminent Discovery

Research Lab 15WD, in charge of researching the safety of the technology, had found an issue with the reaction and particularly SPF912

It was found that under extremely rare circumstances, stalling the reaction would actually turn it into a **miniature black hole** This happened in Test Lab 15WD on January 9th, 2091 Luckily, the reaction was too small to create a black hole that could sustain itself After this discovery, the reactor was immediately shut down and new protocols were implemented

A **Reactor Purge Button** was added to the control room It could be used in the case that the RCC determined that the circumstances were met to create a black hole

The RCC was also equipped with technology to combat the black hole and make it release all of its energy in an explosion rather than imploding and destroying Earth

Please note, that since this event took place, personnel assigned to Lab 15WD have complained of the room being freezing, so much so the laboratory has not been used since.

Incident Report #194729

The reactor was stalled and the conditions were met to cause a black hole Reactor personnel quickly activated the **Power Purge System**, which succeeded

Black Hole Director Firing Control Room Announcement

N/A – Not canon

Incident Report #164729

During Startup, Two employees entered the chamber and were accidentally administered a high dose of radiation almost killing them instantly, they were terminated.

Incident Report #1847291

Employee Code #281 was making an announcement about ensuring safety around the facility Around halfway through the announcement, his ceiling collapsed

Facility 003, Site Aleph

The command and control wing for high-level operatives It contains the site control room, security offices, high-level laboratories, and the main sector and living quadrants for **LEVEL 5 high-ranking officials**, who are required to stay on-site 24/7

IDS

The IDS (Internal Defence systems) are mainly dispersed throughout Site Aleph and various other control points around the facility, this system is state of the art in tracking and firing and can be activated at a moments notice from [REDACTED] [REDACTED]

THIS SECTION IS CLASSIFIED TO ALL LEVEL 5 PERSONNEL

Site aleph is the control area for all **high-ranking officials** and are the heads of site UT. Founded and led by **Astral**, Site aleph has high-level laboratories and living quadrants. And the infamous site control room being able to control EVERYTHING with just the click of a button. Aleph security and **level 7 (SITE COMMAND)** personnel are given a unique weapon, the railgun which is capable of shooting super-sonic rails and obliterating anyone in its path, Hints the name.

Topside Control Room

Connected to Aleph is the Topside control room. Which has many radars, defences, and turret controls.

RC GLASS

On the reactor glass, there is a backup system that uses the same shield material as the reactor to protect the glass in case of a shield explosion or breakage. This prevents reactor operators from falling out of the control room and stops radiation from entering the facility. However, it does have a side effect of giving an unusually non-fatal electric shock to employees who touch it. The reactor control glass shield can protect the glass if the RCC can predict an explosion within 4 seconds.

BUNKER:

The reactor bunker, as simple as it may seem, is an impressive feat of engineering. Due to its close proximity to the reactor, a normal bunker would be unfeasible. The facility bunker is equipped with emergency generators, which, when fueled, provide enough power to operate the hydraulics on the blast doors and maintain the outer containment field.

The outer containment field is an extremely powerful shield, using the same material as the reactor itself, creating a protective force field around the entire bunker. This shield can withstand a direct hit from the combined power of five 50-kiloton nuclear bombs. When activated, the shield draws power from the on-site emergency generators, which then transfer it to the emergency bunker's generator supply.

The Railgun

Aleph security and **level 7 (SITE COMMAND)** personnel are given a unique weapon, the railgun which is capable of shooting super-sonic rails and obliterating anyone in its path, Hints the name.

Da 2 furries that own the group 🦊:

SITE COMMAND // CODENAME: VEX

Codenames: Vex, ZV800, Core533, z3ck200, [REDACTED]

Gender: him/he/any

Species Proton

Description: Not done yet...(Might explode the toaster)

SITE COMMAND // CODENAME: ASTRAL

Name: [REDACTED] [REDACTED] **(ANY STAFF THAT ARE TO SAY THIS INFRONT OF THEM IS TO BE TERMINATED)**

Codenames: Astral, Nexus, [REDACTED]

Gender: They/Them

Species: It is uncertain what species Astral is.

Description: Astral was created from a [REDACTED] [REDACTED] [REDACTED] and still has severe trauma from this day. They are known to be kind and helpful but they *may be chaotic* (NOTE: IS THIS WHY WE HAVE 50 INCIDENT REPORTS FROM THEM?). It's extremely common for Astral to be wearing a suit and top hat. In the back is the *railgun (which im not gonna yap again -astral)*. It was also observed that they were a combat medic, which is why they are skilled with both weapons and medical supplies. Currently, Astral lives in their office and **DOES NOT** want to be revealed to the public unless it's a class 5 emergency.

All other staff files are currently classified.